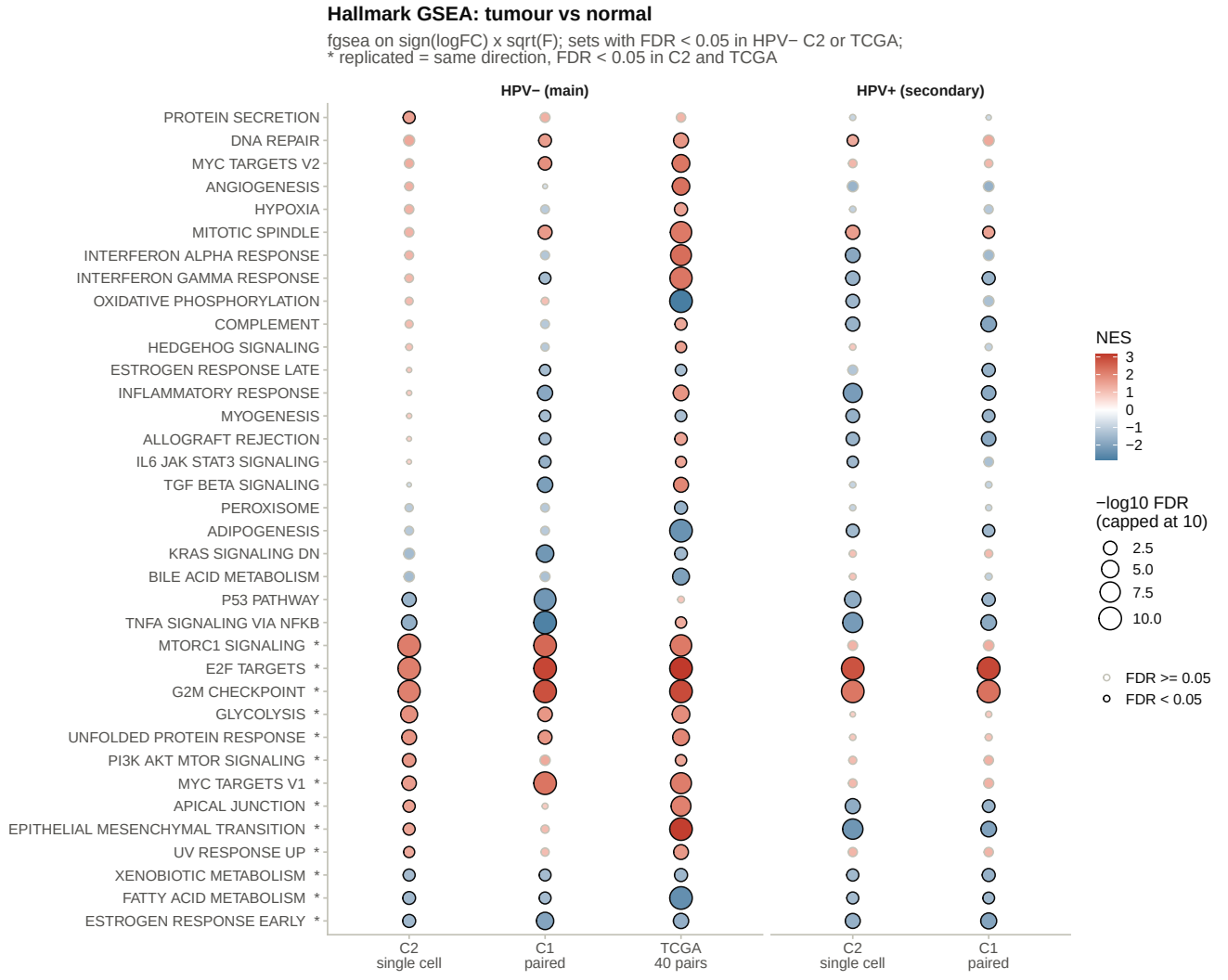
Hallmark GSEA: tumour vs normal
fgsea on sign(logFC) x sqrt(F); sets with FDR < 0.05 in HPV− C2 or TCGA;
* replicated = same direction, FDR < 0.05 in C2 and TCGA
HPV− (main)
HPV+ (secondary)
PROTEIN SECRETION
DNA REPAIR
MYC TARGETS V2
ANGIOGENESIS
HYPOXIA
MITOTIC SPINDLE
INTERFERON ALPHA RESPONSE
INTERFERON GAMMA RESPONSE
OXIDATIVE PHOSPHORYLATION
COMPLEMENT
HEDGEHOG SIGNALING
ESTROGEN RESPONSE LATE
INFLAMMATORY RESPONSE
MYOGENESIS
ALLOGRAFT REJECTION
IL6 JAK STAT3 SIGNALING
TGF BETA SIGNALING
PEROXISOME
ADIPOGENESIS
KRAS SIGNALING DN
BILE ACID METABOLISM
P53 PATHWAY
TNFA SIGNALING VIA NFKB
MTORC1 SIGNALING *
E2F TARGETS *
G2M CHECKPOINT *
GLYCOLYSIS *
UNFOLDED PROTEIN RESPONSE *
PI3K AKT MTOR SIGNALING *
MYC TARGETS V1 *
APICAL JUNCTION *
EPITHELIAL MESENCHYMAL TRANSITION *
UV RESPONSE UP *
XENOBIOTIC METABOLISM *
FATTY ACID METABOLISM *
ESTROGEN RESPONSE EARLY *
C2
single cell
C1
paired
TCGA
40 pairs
C2
single cell
C1
paired
NES
3
2
1
0
−1
−2
−log10 FDR
(capped at 10)
2.5
5.0
7.5
10.0
FDR >= 0.05
FDR < 0.05